Blessed G/Michael Catholic Secondary
First Semester Physics Final Examination for Grade 11

Jan, 2018 E.C

Time Allowed: 2 hr

$\sin 37^0 = \cos 53^0 = 0.6$, $\sin 35^0 = \cos 37^0 = 0.8$, $\sin 20^0 = \cos 70^0 = 0.34$, $\sin 70^0 = \cos 20^0 = 0.94$

$\sin 30^0 = \cos 60^0 = 0.5$, $\sin 60^0 = \cos 30^0 = 0.86$, $g = 10\ m/s^2$

I. *Write "True" if the satmnet is correct or write "False" if the statement is wrong.* *[1 point each]*

false 1. Black hole is a region of space time the gravity of it so strong but light (electromagnetic wave) can escape.

false 2. Once moving at a constant speed, you still have to pedal your bike to keep moving at that speed. This violates Newton's first law of motion.

True 3. If an object is not moving, no resultant force acting upon it.

false (4.) The speed of a freely falling object is proportional to (depends) its mass or weight.

Twe 5. The normal force on an object is **not** always equal in magnitude to the weight of the object. It depends on the orientation of the surface on which an object is placed.

II. *Choose the best answer from the given alternative.* *[1 point each]*

A 6. " The water boiled at a lower temperature when I was visiting the mountains that when I was in other cities with low altitudes." In a scientific method this sentence refer

A. Observation
B. Ask question
C. Research
D. hypothesis

7. *A synthesis of a large body of information that encompasses well-tested and verified hypothesis about certain aspects of the natural world is:*

A. Research
B. hypothesis
C. Conclusion
D. Theory

3 3 → R = 0
3 ← 3 → R = 6
3 120 3 → $9+9-2(9)\cos 60$, $18-9=9$

B 8. Two equal forces acting on a body in different directions have their resultant equal to the magnitude of one of the force. Find the angle between them?

$2^2+2^2+2(2)(2)\cos 120$, $8-8\cos$

A. $\theta = 0^0$ B. $\theta = 120^0$ C. $\theta = 160^0$ D. $\theta = 180^0$

B 9. If two vectors $\vec{A} = 6i - 8j + 4k$ *and* $\vec{B} = 4i - 6j + Pk$ are mutually perpendicular, then the value of p is:

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A. -9 B. -18 C. 4 D. 0

A 10. What does the velocity Vs time graph of a ball falling from a height before hitting the ground?

A. A straight line with a positive slope
B. A parabola
C. A hyperbola
D. A straight line with a negative slope

A 11. Between points A and B, the average speed of a car is 30 m/s, between B and C is 20 m/s, and between C and D is 15 m/s. Calculate the average speed between A and D, if the time taken in the stated sections is the 20s, 10s and 5s, respectively?

A. 25 m/s B. 10 m/s C. 15 m/s D. 30 m/s

C 12. When a body falls under gravity with terminal velocity, what will happen to its average velocity?

A. Average velocity increases
B. Average velocity decreases
C. Average velocity remains constant
D. Average velocity changes exponentially

A 13. The area beneath a velocity-time graph within a given time interval gives

A. The system's displacement during that interval.
B. The distance covered by the system during that interval
C. The system's speed during the interval.
D. The system's acceleration during the interval.

A 14. Addis abeba and Bahir dar are 600 km apart. A small plane leaves Bahir dar traveling toward Addis abeba at an average speed of 90 km/hr. Another plane leaves Addis abeba at the same time, traveling toward Bahir dar, averaging 110 km/hr. How long will it take for them to meet?

A. 3 hr B. 4 hr C. 5 hr D. 6 hr

A 15. If we use plus and minus signs to indicate the directions of velocity and acceleration in one dimension, in which of the following situations may does the object speed up?

A. negative velocity and negative acceleration
B. positive velocity and negative acceleration
C. positive velocity and zero acceleration
D. negative velocity and positive acceleration

16. If you push on a railroad boxcar with a force of 300 N and it doesn't move, you can conclude that

A. Newton's second law is not valid.
B. This force is canceled by its third law partner force.
C. The boxcar has too much mass to accelerate.
D. There must surely be a force of 300 N applied in the opposite direction.

B 17. The 2 measurements necessary for calculating average acceleration are

A. instantaneous velocity & time
B. change in velocity and time
C. distance and speed
D. velocity and distance

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C 18. The displacement of motion is $x = (t^2 - 4t + 10)$ m time is in second. Then the motion is:

A. At rest
B. Uniform motion
C. Uniformly accelerated mtion
D. Uniform circular motion

B 19. Suppose you are in a car going due east on a straight highway at a constant speed of 30 km/hr. Which of the following is NOT true?

A. You and the car's distance is constantly changing
B. Your direction is constantly changing
C. Your speed is constant
D. Your velocity is constant

20. It is easier to pull a rolling bag on a road than to push. This is because when the rolling bag is pulled,

A. the vertical component of the force acts in the direction of gravity and weight thereby weight of the roller is increased and friction increased.

B. the vertical component of the force acts in opposite direction to the weight, thereby weight of the roller is decreased and friction decreased.

C. the premise is wrong, because pulling a roller cannot be easier than pushing.

D. Has no enough information to deside answer



21. South still displays with great pride the large sign boasting of their 1996 State Championship Girl's Basketball Team. The 60 kg sign hangs from two cables which make an angle of 37° with the horizont Determine the tension in each of the cables.

A. 250 N B. 500 N C. 600 N D. 1200 N

22. Dawit and Solomon have tied a rope to Matthew's sled and are momentarily pulling him through the snow. Dawit is pulling with force A and Mom with force B. Resultant force is?

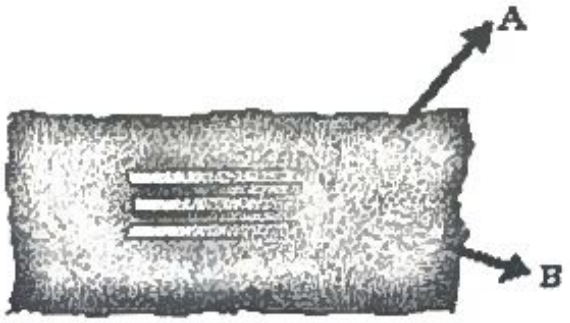
Force A: 50 N at 37° north of east and force B: 60 N at 30° south of east

A. 92 N at 30° south of east

B. 109 N at 30° south of east

C. 92 N to wards East

D. 109 N to wardes East



23. The study of motion of an object by considering the cause to change state of motion of an object is:

A. Dynamics B. kinematics C. Mechanics D. Optics

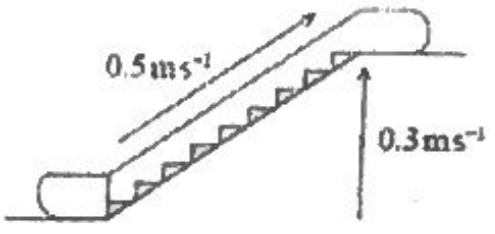
24. The steps on an escalator move with a speed of 0.5 m/s^2. The vertical component of their velocity is 0.3m/s^2 upwards as shown in the figure below. Which of the following is the horizontal component of velocity for the escalator steps?

A. 0.15 m/s^2

B. 0.2 m/s^2

C. 0.4 m/s^2

D. 0.8 m/s^2



25. The motion of a block sliding down a frictionless ramp can be described as motion with

A. a constant speed, independent of the slope of the ramp.

B. a constant speed that depends on the slope of the ramp.

C. a constant acceleration less than 10 m/s^2.

D. a constant acceleration greater than 10 m/s^2.

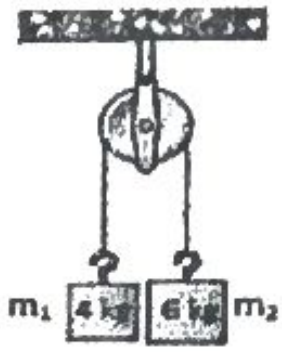
B 26. Two masses ($m_1 = 4kg$ and $m_2 = 6kg$), are connected by a light cord that passes over a massless, frictionless pulley as shown in Figure below. This arrangement is called Atwood's machine and sometimes is used to measure the acceleration due to gravity. Find the magnitude of acceleration of the two masses and the tension in the cord respectively?

A. 0.6 m/s^2 and 20 N

B. 2 m/s^2 and 48 N

C. 3 m/s^2 and 20 N

D. 3 m/s^2 and 48 N



D 27. Which one of the following is NOT correct according to Newton's 1st law

A. May be at rest

B. May be uniform motion

C. no net force

D. Non zero acceleration

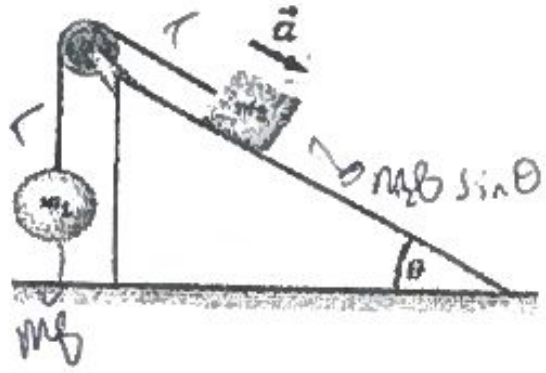
A 28. A ball of mass m_1 and a block of mass m_2 are attached by a lightweight cord that passes over a frictionless pulley of negligible mass, as in Figure below. The block lies on a frictionless incline of angle θ. Find the magnitude of the acceleration of the two objects in terms of m_1, m_2, g and θ?

A. $\frac{m_1 g - m_2 g \sin\theta}{m_1 + m_2}$

B. $\frac{m_1 g - m_2 g \cos\theta}{m_1 + m_2}$

C. $\frac{m_2 g \cos\theta - m_1 g}{m_1 + m_2}$

D. $\frac{m_2 g \sin\theta - m_1 g}{m_1 + m_2}$



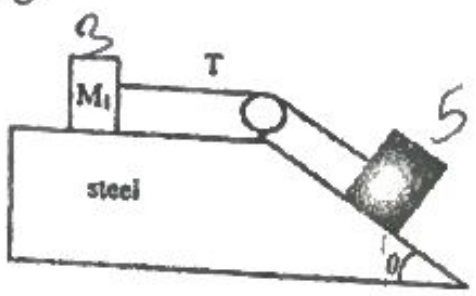
B 29. An aluminum block of mass $m_1 = 3$ kg and a copper block of mass $m = 5$ kg are connected by a light string over a friction less pulley. They sit on a steel surface as shown in the figure, where $\theta = 53°$. The coefficient of kinetic friction between the steel surface and each block is 0.3. If the blocks are released from rest what is the tension T of the string?

A. 29.4N

B. 17.25N

C. 12.25N

D. 8.6N



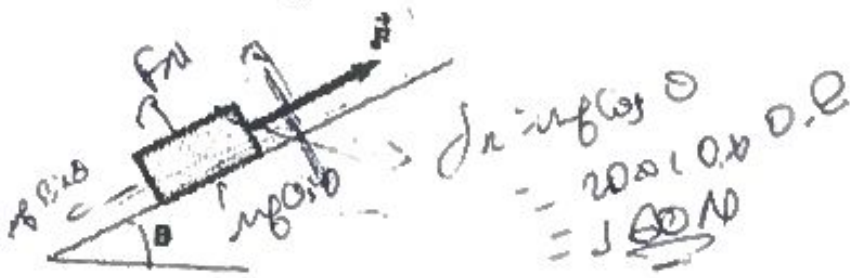
B 30. A 20 kg block is placed on a smooth inclined plane shown in Figure below. What is the normal force? Take the value coefiecient of friction μ=0.4, force $F = 40\ N$ and $\theta = 37^0$

A. 120 N

B. 160 N

C. 200 N

D. 960 N



A 31. A 25 kg block is initially at rest on a horizontal surface. A horizontal force of 75 N is required to set the block in motion. After it is in motion, a horizontal force of 60 N is required to keep the block moving with constant speed. Find the coefficients of static and kinetic friction from this information respectively?

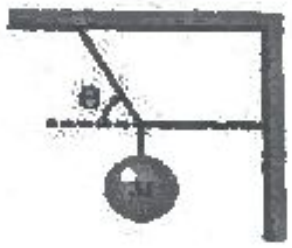
A. 0.3 and 0.24 B. 0.06 and 0.1 C. 0.25 and 0.2 D. 0.4 and 0.32

C 32. Ten seconds after starting from rest, an object falling freely downward on Earth will have a speed of about

A. 10 m/s B. 50 m/s C. 100 m/s D. more than 100 m/s

A 33. Mass m= 8 kg is supported by two strings is in equilibrium as shown in Figure below. Find the tension in the horizontal string. Take $\theta = 53°$?

A. 60 N
B. 80 N
C. 100 N
D. 120 N



III. *Fill in the blank space. [4 point total]*

34. A unit vector vector along the x-axis is denoted by $\hat{i}$, that along the y-axis is denoted by $\hat{j}$ and along the z-axis is denoted by $\hat{k}$.

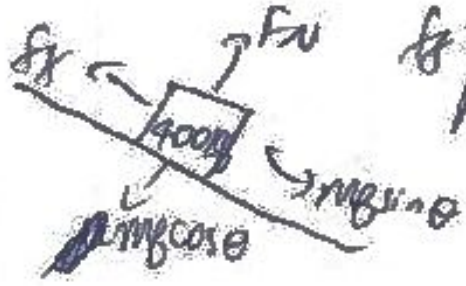
35. Centripetal $\vec{a}$ is an acceleration in circular motion caused by the change in the direction of velocity. Its direction is always towards the center of the circle.

36. Normal force is a force with which to objects push one another and acting perpendicularly at the surface of contact

IV. *Work out (Write only final answer). [4 point total]*

If a 400 kg block is at rest on an incline without slipping, then friction must equal the component of the weight of the object parallel to the incline. this requires greater and greater friction for steeper slopes. Coefficient of friction 0.2. take $\theta = 37^0$





$f = \mu mg\sin\theta$
$\mu mg\cos\theta = mg\sin\theta$
$\mu = \frac{\sin\theta}{\cos\theta}$
$\mu = \tan\theta$
$\mu = \tan 37° = \frac{\sin 37°}{\cos 37°}$
$= \frac{0.6}{0.8} = \frac{6}{8} = \frac{3}{4}$
$= 0.75$

37. Draw the free body diagram

38. This angle is called the critical angle. For what angle would the block begin to slide if the surface were frictionless?

$\theta = \tan(\mu) = \tan^{-1}(0.2) = 11$

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ANSWER SHEET

Name: Abubakar Menale Grade: 11 Section: C R.No 33

1. False	8. B	15. A D	22. C	29. B
2. False	9. B	16. C D	23. A	30. B
3. True	10. A	17. B	24. C	31. A
4. False	11. A	18. C	25. C	32. C
5. True	12. C	19. B	26. B	33. A
6. A	13. A	20. B	27. D	
7. D	14. A	21. B	28. A D	

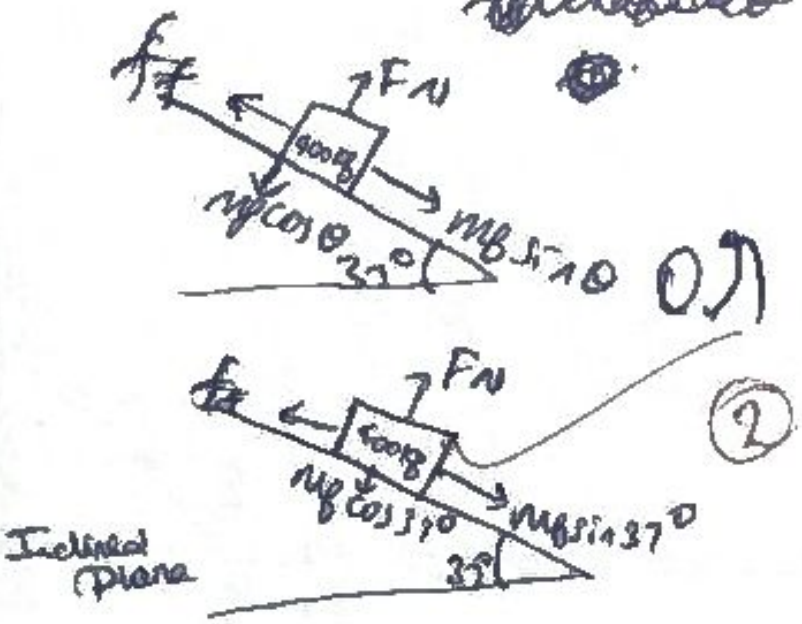
34. $\hat{i}$, $\hat{j}$ and $\hat{k}$

35. Centripetal acceleration

36. Normal force

37. f_s = frictional force

Incorrect



(2)

38. ~~$\mu_s = \tan\theta_c$~~

~~$\theta_c = \tan^{-1}(\mu_s)$~~

~~$= \tan^{-1}(0.2)$~~ ~~for each D~~

~~But for friction less, it will slide at any angle.~~

→ If the inclined plane were friction less the block would slide at any angle since $\mu = 0$. Making until reach 90 so θ can be any angle.

(2)

final answer - θ can be any angle

please show mathematically